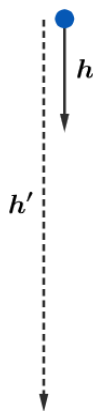


2007 F=ma Exam: Problem 4

Kevin S. Huang



Recall one of our kinematics equations for uniform acceleration:

$$x = v_0 t + \frac{1}{2} a t^2$$

For dropping an object from rest, we have $v_0 = 0$ and $a = g$:

$$d = \frac{1}{2} g t^2$$

In the first unit of time,

$$h = \frac{1}{2} g t_0^2$$

In the total two units of time,

$$h' = \frac{1}{2} g (2t_0)^2$$

Hence, the object fell

$$h' - h = \frac{1}{2} g (2t_0)^2 - \frac{1}{2} g t_0^2 = 3 \left(\frac{1}{2} g t_0^2 \right) = 3h$$

during the second unit of time so the answer is C.